1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights that tumor transcriptomes and specific circRNAs (e.g., circMMP9) associate with cisplatin response, yet some tumors with favorable microenvironmental features (e.g., immune infiltration) still fail to respond. Why does a transcriptional/post-transcriptional program override microenvironmental predictors of cisplatin sensitivity?

Working Hypothesis

Primary hypothesis: In bladder cancer, high eIF4E-driven cap-dependent translation increases circMMP9 biogenesis, and this circMMP9 promotes tumor-intrinsic resistance to cisplatin, effectively diminishing the predictive value of tumor-infiltrating lymphocytes (TILs). Consequently, tumors with high eIF4E and high circMMP9 will exhibit cisplatin resistance irrespective of high TIL density; disrupting circMMP9 or reducing eIF4E activity will restore cisplatin sensitivity. Secondary hypotheses: (i) circMMP9 acts as a sponge for miRNAs that regulate DNA damage response and apoptotic pathways, leading to attenuated cisplatin-induced DNA damage signaling; (ii) EIF4A3 and other splicing back-end factors modulate circMMP9 biogenesis and are necessary for the full resistant phenotype; (iii) circTRAM1 and related modifiers can modulate the axis but do not dominate the cisplatin response when eIF4E/circMMP9 are high; (iv) combining circMMP9 inhibition with cisplatin will have synergistic anti-tumor effects and better clinical predictivity than TIL metrics alone.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	eIF4E, circMMP9
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

eIF4E: eIF4E-driven modulation of circMMP9).

circMMP9: circMMP9) associate with cisplatin response, yet some tumors with favorable microenvironmental features (e.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	HRG1	Supporting modifier from knowledge graph

Secondary	EIF4A3	Supporting modifier from knowledge graph
Secondary	circZFAND6	Supporting modifier from knowledge graph
Secondary	circTRAM1	Supporting modifier from knowledge graph
Secondary	DEFB103	Supporting modifier from knowledge graph
Secondary	CENPF	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: manipulate eIF4E and circMMP9 in bladder cancer cell lines and measure cisplatin sensitivity (IC

Prediction 2: , apoptosis, and DNA damage signaling;

Prediction 3: disrupt circMMP9 using antisense oligonucleotides or CRISPR-based back-splice disruption; assess rescue of cisplatin sensitivity;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- HRG1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- EIF4A3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.